

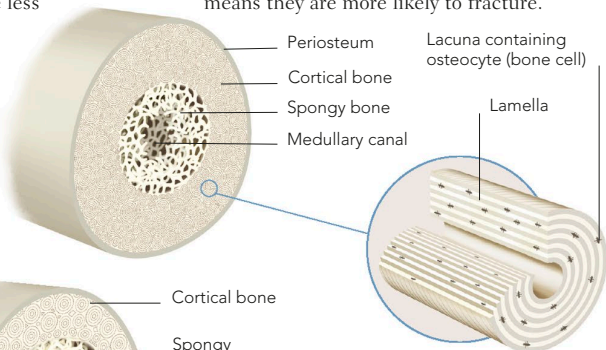
OSTEOPOROSIS

For bones to stay healthy, bone tissue is continually being broken down and replaced. Sex hormones are essential for this process. With the decline in production of sex hormones in both sexes after middle age, bones become thinner and more porous. Oestrogen levels fall rapidly in women after the menopause, which can lead to severe thinning, or osteoporosis. The decline in testosterone in men is gradual and, in general, males are less

prone to osteoporosis. Exercise is essential for maintaining bone health, and a lack of activity is a predisposing factor to developing osteoporosis. Other factors influencing the development of osteoporosis include smoking, corticosteroid treatment, rheumatoid arthritis, an overactive thyroid, and long-term kidney failure. The decreased density of osteoporotic bones means they are more likely to fracture.

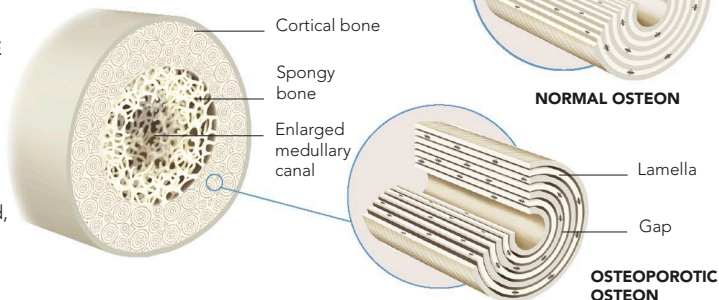
STRUCTURE OF NORMAL BONE

The outer periosteum encloses a band of hard cortical bone. Within this is a layer of spongy, or cancellous, bone. Hard bone is composed of osteons, which are tightly packed, concentric layers (lamellae) formed by osteocytes.



STRUCTURE OF OSTEOPOROTIC BONE

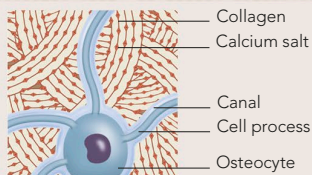
The mineral density (mainly calcium and phosphorus) is reduced from two-thirds to one-third. The medullary canal through the bone's centre is enlarged, while gaps between the lamellae contribute to the fragility of the bone.



WHY OSTEOPOROSIS OCCURS

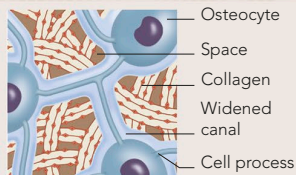
Bone tissue is built up by the deposition of minerals (mainly calcium salts) on a framework of collagen fibres. It is continually broken down and rebuilt in order to allow growth and repair.

Osteoporosis develops when the rate at which fibres, minerals, and cells are broken down becomes much greater than the formation of new tissue.



NORMAL BONE

Osteocytes form collagen fibres and aid calcium deposition. Calcium moves in canals between bone and blood in response to hormones.



OSTEOPOROTIC BONE

In osteoporosis, collagen and minerals are broken down faster than they form. Canals widen, new spaces appear, and bone weakens.